

# ANANTHA CHARY C M

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## EDUCATION

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| <b>B.E. in Information Science</b><br>BMS College of Engineering, Bengaluru<br>CGPA 9.54 | 2023–Present |
| <b>ST NORBERT PU College</b><br>PU (93.16%)                                              | 2021–2023    |
| <b>JNANAVAHINI English Medium High School</b><br>SSLC (97.28%)                           | 2020–2021    |

## EXPERIENCE

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| <b>PROJECT INTERN (CPP) – HEWLETT PACKARD ENTERPRISE</b>  <a href="#">GitHub</a> <i>Feb 2026 – Jul 2026</i>                                                                                                                                                                                                                                                                                                                                                              |
| <ul style="list-style-type: none"><li>Built a self-organizing RAG framework that continuously monitors its own retrieval quality, using a rule-based detection engine to catch failing queries and automatically diagnose the root cause.</li><li>Designed a 4-stage repair cascade resolving 82% of flagged failures with atomic snapshot-based rollback, adding only ~48 ms of user-facing latency via a fully asynchronous background worker.</li></ul> <b>Tech &amp; Tools:</b> Python, FastAPI, LangChain, Streamlit, Pinecone, SQLite, Docker, Ollama |

## PROJECTS

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| <b>Resilience Grid – AI-Driven Energy Supply Chain Resilience for Import-Dependent Economies</b>  <a href="#">GitHub</a>                                                                                                                                                                                                                                                                                                                                                                                                    |
| <ul style="list-style-type: none"><li>Built a platform that tracks risk across India's key import corridors in real time, combining geopolitical events, sanctions data, vessel movements, and commodity prices into a single live risk score, with alerts triggered whenever conditions change.</li><li>Added a geospatial digital twin to simulate disruption scenarios, plus an optimizer that determines optimal strategic-reserve drawdowns under uncertainty, stress-tested across dozens of combinations of shocks.</li></ul> <b>Tech &amp; Tools:</b> Python, FastAPI, PuLP, React, WebSockets, Ollama |
| <b>EduRAG: Local Multimodal Q&amp;A and Self-Assessment Platform</b>  <a href="#">GitHub</a>                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <ul style="list-style-type: none"><li>Developed a local RAG pipeline for parsing PDFs, slides, and documents, using a vision model to describe diagrams and PyMuPDF to extract tables.</li><li>Integrated Pinecone, FlashRank re-ranking, and Jaccard-based answer verification.</li></ul> <b>Tech &amp; Tools:</b> Python, Streamlit, LangChain, Pinecone, Ollama                                                                                                                                                                                                                                             |
| <b>SpaceSim – Solar System &amp; Artemis II Mission Simulator</b>  <a href="#">GitHub</a>                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <ul style="list-style-type: none"><li>Built an N-body Solar System and Artemis II lunar flyby simulation with velocity Verlet integration, multi-stage rocket modeling, and real-time 3D rendering via Genesis.</li><li>Streamed live telemetry over WebSockets to a Three.js web viewer with an interactive mission HUD.</li></ul> <b>Tech &amp; Tools:</b> Python, Genesis, NumPy, WebSockets                                                                                                                                                                                                                |

## TECHNICAL SKILLS

- Languages:** Python, Java, C++
- Frameworks & Libraries:** React, Streamlit, scikit-learn, YOLOv8, OpenCV
- Backend & Databases:** FastAPI, Flask, Node.js, MySQL, MongoDB, PostgreSQL
- GenAI & RAG:** RAG Pipelines, LangChain, Ollama, Pinecone, Embeddings
- Developer & Cloud Tools:** Git, GitHub, Docker, AWS (EC2, S3), Google Cloud (Beginner)

## PUBLICATIONS

- Performance Evaluation of TCP Congestion Control Variants Across Application Workloads in Cloud Based Networks, **Scientific Reports**, 2026. DOI: 10.1038/s41598-026-61100-w

## CERTIFICATIONS

- Google Cloud Engineering Certificate** – Google Skills
- Full Stack Web Development** – Udemy

## ACHIEVEMENTS

- HyperAPI Global Hackathon – Winner**  
Built an end-to-end pipeline for processing financial documents and detecting anomalies using HyperAPI.
- Roll,Commit,Ship!! – Winner**  
Built a gamified, retro-themed civic crowdfunding platform using React, Flask, and PostgreSQL.